

The quadratic equation and its roots

The first equation of the course whose graph is not a line — and the first that can have two answers, one, or none at all.

LESSON 62–63

2 × 40 MIN

ALGEBRA 8, CHAPTER III \$20

STAGE 9 · BEYOND

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Recognise a quadratic equation and name its coefficients.
- Check whether a given number is a root.
- Solve a quadratic by factorising.
- Read the number of roots off the graph.

TEXTBOOK

National	Chapter III, pp. 122–128
Cambridge	Extension beyond Stage 9
Workbook	—

01

Explanation

What a quadratic equation is

DEFINITION

An equation of the form

$$ax^2 + bx + c = 0, \quad a \neq 0$$

is a **quadratic equation**. Here a is the **leading coefficient**, b the second coefficient and c the **free term**.

The condition $a \neq 0$ matters: without it the x^2 disappears and the equation is linear.

EQUATION	A	B	C
$x^2 - 5x + 6 = 0$	1	-5	6
$3x^2 + 2x - 8 = 0$	3	2	-8
$2x^2 - 7 = 0$	2	0	-7
$x^2 + 4x = 0$	1	4	0

Write the equation in this standard order **before** reading off the coefficients — and mind the signs: in $x^2 - 5x + 6$ the coefficient b is -5 , not 5 .

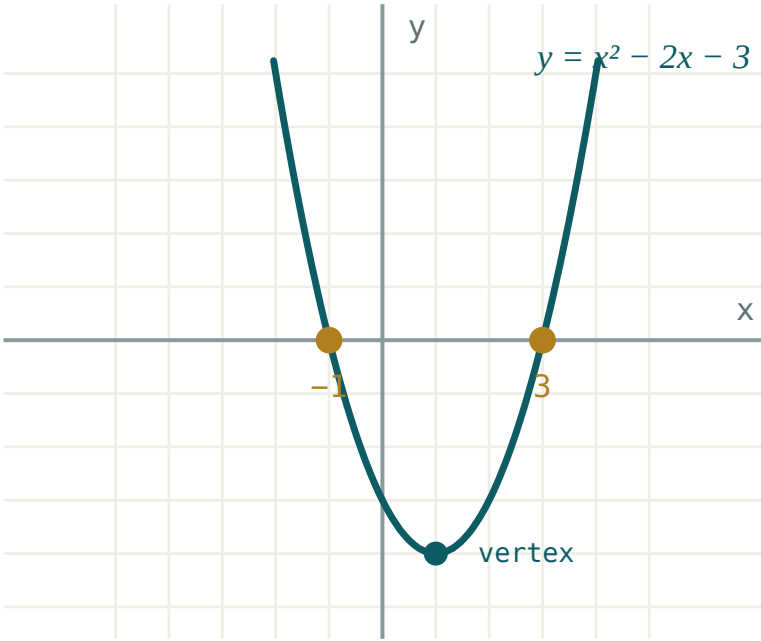
Roots

DEFINITION

A **root** of the equation is a value of x that makes it true.

To check, substitute. Is 3 a root of $x^2 - 5x + 6 = 0$? $9 - 15 + 6 = 0$ ✓ — yes.

Graphically, the roots are where the parabola $y = ax^2 + bx + c$ crosses the x -axis.



The graph of $y = x^2 - 2x - 3$ crosses the x -axis at -1 and 3 — the two roots of $x^2 - 2x - 3 = 0$.

A parabola can cross the axis twice, touch it once, or miss it entirely — so a quadratic equation has **two roots, one root, or none**. That is the single biggest difference from the linear equations of Grade 7.

Solving by factorising

THE ZERO PRODUCT RULE

If $A \cdot B = 0$ then $A = 0$ or $B = 0$. Nothing else can happen — this is the only reason factorising solves equations.

So write the quadratic as a product equal to zero, then set each bracket to zero in turn:

$$x^2 - 5x + 6 = 0$$

$$(x - 2)(x - 3) = 0$$

$$x - 2 = 0 \text{ or } x - 3 = 0 \implies x = 2 \text{ or } x = 3$$

To factorise $x^2 + px + q$, find two numbers with **product** q and **sum** p . For $x^2 - 5x + 6$: product 6, sum $-5 \rightarrow -2$ and -3 .

THE RULE NEEDS A ZERO ON THE RIGHT

From $(x - 1)(x - 4) = 6$ you may **not** write $x - 1 = 6$. Expand, move everything to one side, and factorise again.

02 Worked examples

EXAMPLE 1 Solve $x^2 - 7x + 12 = 0$

- 1 Two numbers with product 12 and sum -7 .
Both negative, since the product is positive and the sum negative.
- 2 -3 and -4
- 3 $(x - 3)(x - 4) = 0$
- 4 $x = 3$ or $x = 4$
Check: $9 - 21 + 12 = 0$ ✓

ANSWER

$$x = 3, x = 4$$

EXAMPLE 2 Solve $x^2 + 2x - 15 = 0$

- 1 Product -15 , sum 2.
Opposite signs, since the product is negative.
- 2 5 and -3
- 3 $(x + 5)(x - 3) = 0$
- 4 $x = -5$ or $x = 3$

ANSWER

$$x = -5, x = 3$$

EXAMPLE 3 *Is $x = -2$ a root of $2x^2 + 3x - 2 = 0$?*

① $2(-2)^2 + 3(-2) - 2$
Substitute.

② $= 8 - 6 - 2$
Mind the sign: $(-2)^2 = 4$, not -4 .

③ $= 0$
It works.

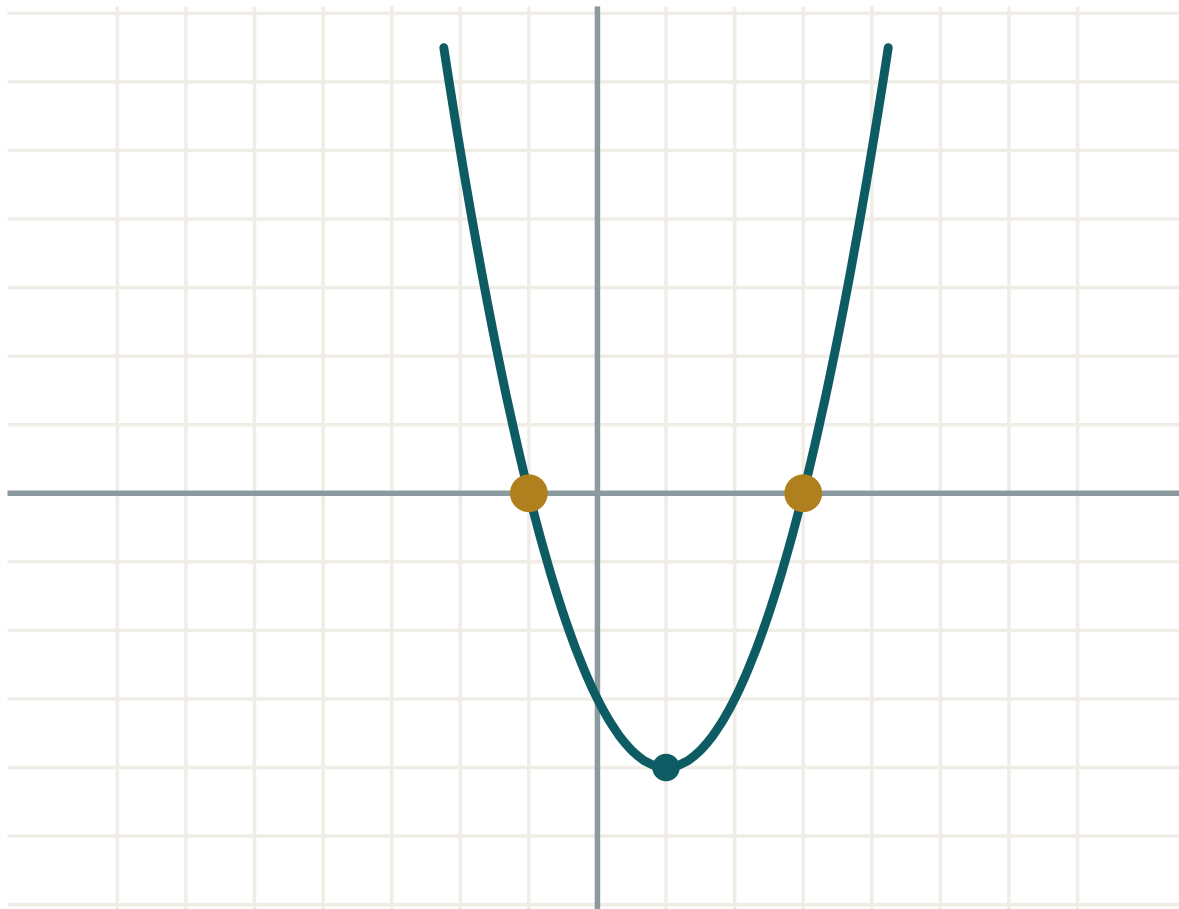
ANSWER

Yes, $x = -2$ is a root.

03 Interactive model

Set $a = 1$, $b = -2$, $c = -3$ and read the roots off the graph before doing any algebra.

Roots and the discriminant

a **1**b **-2**c **-3** $D = b^2 - 4ac$ **16**roots **two** x_1 **-1** x_2 **3** $x_1 + x_2$ **2** $x_1 \cdot x_2$ **-3**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Quadratic equation	Kvadrat tenglama	Квадратное уравнение
Root of an equation	Tenglamaning ildizi	Корень уравнения
Coefficient	Koeffitsiyent	Коэффициент
Leading coefficient	Bosh koeffitsiyent	Старший коэффициент
Free term (constant)	Ozod had	Свободный член
Factorise	Ko'paytuvchilarga ajratish	Разложить на множители
Zero product rule	Ko'paytmaning nolga tengligi	Произведение равно нулю
Parabola	Parabola	Парабола
Substitute and check	O'rniga qo'yib tekshirish	Подставить и проверить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

In $3x^2 - 5x + 2 = 0$, the coefficient b is:

A quadratic equation can have:

$(x - 2)(x + 5) = 0$ gives:

$x^2 - 5x + 6 = 0$ factorises to:

$$(x - 2)(x - 3)$$

$$(x + 2)(x + 3)$$

$$(x - 1)(x - 6)$$

$$(x - 2)(x + 3)$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Name a, b, c in $x^2 + 3x - 4 = 0$

ANSWER $1, 3, -4$

2. Name a, b, c in $2x^2 - 5 = 0$

ANSWER $2, 0, -5$

3. Is $x = 2$ a root of $x^2 - 4 = 0$?

ANSWER Yes.

4. Is $x = 1$ a root of $x^2 + x - 2 = 0$?

ANSWER Yes.

5. Solve $(x - 3)(x + 1) = 0$

ANSWER $x = 3, x = -1$

6. Solve $(x + 4)(x + 2) = 0$

ANSWER $x = -4, x = -2$

7. Solve $x(x - 5) = 0$

ANSWER $x = 0, x = 5$

Medium · the standard question

7 PROBLEMS

1. Solve $x^2 - 7x + 12 = 0$

ANSWER $x = 3, x = 4$

2. Solve $x^2 + 2x - 15 = 0$

ANSWER $x = -5, x = 3$

3. Solve $x^2 - x - 6 = 0$

ANSWER $x = 3, x = -2$

4. Solve $x^2 + 7x + 10 = 0$

ANSWER $x = -2, x = -5$

5. Solve $x^2 - 9x + 20 = 0$

ANSWER $x = 4, x = 5$

6. Is $x = -2$ a root of $2x^2 + 3x - 2 = 0$?

ANSWER Yes.

7. Solve $x^2 - 2x - 8 = 0$

ANSWER $x = 4, x = -2$

Hard · stretch and reason

7 PROBLEMS

1. Solve $2x^2 - 5x + 3 = 0$ by factorising

ANSWER $(2x - 3)(x - 1) = 0$, so $x = 1.5, x = 1$

2. Solve $3x^2 + 5x - 2 = 0$

ANSWER $(3x - 1)(x + 2) = 0$, so $x = \frac{1}{3}, x = -2$

3. Solve $(x - 1)(x - 4) = 6$

ANSWER $x^2 - 5x - 2 = 0$ — does not factorise nicely; the roots come from the formula next lesson.

4. One root of $x^2 + bx - 6 = 0$ is 2. Find b and the other root.

ANSWER $4 + 2b - 6 = 0$ gives $b = 1$; the other root is -3 .

5. For which c does $x^2 - 6x + c = 0$ have $x = 2$ as a root?

ANSWER $4 - 12 + c = 0$, so $c = 8$

6. Solve $x^2 = 5x$

ANSWER $x(x - 5) = 0$: $x = 0$ or $x = 5$. Never divide by x — you would lose the root 0.

7. Write a quadratic equation whose roots are 3 and -7 .

ANSWER $(x - 3)(x + 7) = 0$, i.e. $x^2 + 4x - 21 = 0$

07 Homework

Homework — 6 problems

Algebra 8, Chapter III, pp. 122–128. Check every root by substituting.

1. Name a, b, c in $4x^2 - 7x + 1 = 0$
2. Solve $x^2 - 8x + 15 = 0$
3. Solve $x^2 + 5x - 14 = 0$
4. Solve $x^2 = 7x$

5. Is $x = -3$ a root of $x^2 + x - 6 = 0$?
6. Write a quadratic equation with roots 2 and -5